

4	(a)	Stationary waves are waves whose <u>waveform does not advance</u> and there are <u>no translation of energy</u> .	B1 B1
	(b)	(i)	
		The sound wave produced by the loudspeaker travels down the pipe and gets <u>reflected at the piston</u> .	B1
		The incident wave and reflected wave have the <u>same frequency and speed</u> and are <u>travelling in opposite directions</u> hence	B1
		(ii)	
		Superpose/meet to form a stationary wave.	A0
		Wavelength of stationary wave = 0.24 m	C1
		$f = v / \lambda$ $= 330 / 0.24$ $= 1375 \text{ (1375 Hz)}$	M1 A1